

TRENTOS-M Systems on zynqmp

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Modified Components/Libs

Component	Change commit	Merged (✓✗)	Note
uart	5c65b45b362 b8040d607ad	✓	Adding plat support files Adapting printf modifiers for the 64-bit arch
test_uart	38ed7dce8fe	✓	Adding plat support files
TimeServer	904d6f231a8	✗ (SEOS-000_zynqmp_support branch)	Adapting printf modifiers for the 64-bit arch - not merged because the proper printf modifier macro is only available in the new sel4 version.
sdk-sel4-camkes	cf39bda32e4	✗ (SEOS-000_zynqmp_support branch)	New sel4 version (kernel and libs) which (among other things) fixes some architecture assumptions which caused the printf modifier errors in <code>timeserver</code> and <code>test_timeserver</code>
chanmux	8d9fd2a0951	✓	Adding plat support files
chanmux_nic_driver	ea69a528eb3	✓	Fixing <code>printf</code> formatting issues for 64-bit architecture
os_logger	85784eec95b	✓	Fixing <code>printf</code> formatting issues for 64-bit architecture
demo_iot_app	f6948bc4e8b	✓	Fixing <code>printf</code> formatting issues for 64-bit architecture
StoragSserv er	93c0e052d4a	✗ (SEOS-1618 branch)	Adapting printf modifiers for the 64-bit arch - not merged because the proper printf modifier macro is only available in the new sel4 version.

CI Support

Because of the fact that the `xilinx-qemu` is needed for booting `zynqmp` in QEMU, which is only available in a modified container, the supported test systems are executed for `zynqmp` in CI in a branch: https://jenkins.devnet.hensoldtcyber/job/trentos-m-sdk/job/SEOS-000_zynqmp_support/. Currently, the following systems are supported:

Test system	Note
test_demo_hello_world	
test_timeserver	
test_uart	
test_chanmux	
test_crypto_api	
iot_demo_app	
test_network_api	Builds, but does not work (see WiP)

Work in Progress

Ethernet support

This section will describe the work done in the scope of [SEOS-2752](#) - Getting issue details... STATUS which is currently placed on hold.

As the base for the zynqmp ethernet support, we used the ethernet driver provided by DornerWorks. The original package containing the ethernet driver together with other modifications (the scope of the modifications is not fully clear, but is not limited to the ethernet support nor only to the zynqmp platform) is found here: https://bitbucket.hensoldt-cyber.systems/projects/DW/repos/seL4_hypervisor/browse. This repository is a "meta-repository" that is based on <https://github.com/seL4/camkes-arm-vm-manifest> and collects the source code modified by DW relevant for zynqmp porting (the original DW archive contained a couple of repositories not part of this project that are found alongside other DW repositories here <https://bitbucket.hensoldt-cyber.systems/projects/DW> as well as binaries related to virtualisation that are in the original archive on the NAS: /Public/Deliverables/DornerWorks).

As a test system to build and test the driver functionality, the **test_network_api** systems is used. Instead of using the NIC_ChanMux as for the zynq7000 platform, the zynqmp ethernet driver is used together with the NIC_iMX6 component which is modified to support the additional target. Also, the test system is modified to accommodate the new target. The overview of the made changes is described in the table below:

Component	Change commit	Merged (✓✗)	Note
NIC_iMX6	f82133b8d9c e6f39bc06d7	✗	Adding support for zynqmp: new plat directory containing zynqmp specific camkes files
test_network_api	14342e1abc5	✗	Adding support for zynqmp: new plat directory containing zynqmp specific camkes files
SysLogger	cff69a12576	✗	Fixing printf formatting issues for 64-bit architecture
seL4_util_libs	4ce19592b0e bd8f68e9e7b	✗	Adding ethernet driver files from the DW repository: https://bitbucket.hensoldt-cyber.systems/projects/DW/repos/util_libs/browse Adapting the drivers to the new seL4 version/SDK, specifically to the new driver architecture described here: https://bitbucket.hensoldt-cyber.systems/projects/EXT/repos/util_libs/browse/libplatsupport/docs/driver_environment/index.md

Finally, the entire branch is based on the [SEOS-1695](#) - Getting issue details... STATUS branch since it contains modifications for the NIC_iMX6 QEMU functionalities that may be useful for zynqmp porting as well.

Current state and issues

The current state of the test_network_api system for the zynqmp platform is as follows:

- The system builds successfully
- When started the system does not initialize correctly and immediately throws an error when trying to find the ethernet driver. Based on short investigation it seems that the function **ethif_zynqmp_init_module** (seL4_util_libs/libethdrivers/src/plat/zynqmp/zynqmp.c) is not called from **server_init** (in NIC_iMX6) which might be caused by:
 - the fact that the NIC_iMX6 component lacks modifications that would allow it to correctly find the zynqmp ethernet driver
 - zynqmp ethernet driver is not correctly adapted to the driver structure explained here: https://bitbucket.hensoldt-cyber.systems/projects/EXT/repos/util_libs/browse/libplatsupport/docs/driver_environment/index.md
 - test_network_api is not correctly modified
- Before debugging the driver itself, it would be beneficial to double-check the system is set up correctly, i.e., that the zynqmp platform is "correctly added"

test_network_api output

```
server_init@ethdriver.c:614 Unable to find an ethernet device, code -2
pre_init@seL4SingleThreadedComponent.template.c:59 Failed to initialize 0x43c5d0
INFO: /host/source/src/tests/test_network_api/components/ConfigServer/src/ConfigServer.c:51: Starting
ConfigServer...
INFO: /host/source/src/tests/test_network_api/components/ConfigServer/src/ConfigServer.c:27: ConfigServer:
Initializing with mem backend...
INFO: /host/source/src/tests/test_network_api/components/ConfigServer/src/ConfigServer.c:58: Config Server
initialized.
INFO: /host/source/src/tests/test_network_api/components/Ticker/Ticker.c:15: ticker running
[nwApp2] INFO: /host/source/src/tests/test_network_api/components/TestAppServer/test_app_server.c:65:
Starting test_app_server...
[nwStack2] INFO: /host/source/src/tests/test_network_api/components/NwStack/network_stack.c:147: [NwStack
'nwStack2'] starting
[nwStack2] INFO: /host/source/src/tests/test_network_api/components/NwStack/network_stack.c:221: [NwStack
'nwStack2'] IP ADDR: 10.0.0.11
[nwStack2] INFO: /host/source/src/tests/test_network_api/components/NwStack/network_stack.c:222: [NwStack
'nwStack2'] GATEWAY ADDR: 10.0.0.1
[nwStack2] INFO: /host/source/src/tests/test_network_api/components/NwStack/network_stack.c:223: [NwStack
'nwStack2'] SUBNETMASK: 255.255.255.0
[nwStack2] INFO: /host/OS-SDK/pkg/libs/os_network_stack/network_core/src/network_stack_pico_nic.c:219: MAC:
00:00:00:00:00:00
[nwStack2] INFO: /host/OS-SDK/pkg/libs/os_network_stack/network_core/src/network_stack_pico_nic.c:230:
PicoTCP Device created: trentos_nic_driver
[nwStack2] INFO: /host/OS-SDK/pkg/libs/os_network_stack/network_core/src/network_stack_pico.c:407: [socket 0
/0x6c6920] created new socket
[nwStack2] INFO: /host/OS-SDK/pkg/libs/os_network_stack/network_core/src/network_stack_pico.c:522: [socket 0
/0x6c6920] binding to port 5555
[nwApp2] INFO: /host/source/src/tests/test_network_api/components/TestAppServer/test_app_server.c:91:
launching echo server
nic_rpc_rx_data@ethdriver.c:297 Device not initialized
nic_rpc_rx_data@ethdriver.c:297 Device not initialized
nic_rpc_rx_data@ethdriver.c:297 Device not initialized
```